

Effect of generative artificial intelligence on academic publishing ethics: The role of user agreements

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Abstract

Generative artificial intelligence (AI) such as ChatGPT are rapidly expanding in scientific research and publication due to their excellent language understanding, generation and knowledge reasoning capabilities, but the accompanying ethical issues of publication cannot be ignored. User agreements can provide behavioural guidance for users to use generative AI tools, but its ethical guiding role in academic publishing is unclear. This study aims to investigate the ethical implications of generative AI in academic publishing, with a specific focus on the role of user agreements. We selected 98 large language models (LLMs) which were divided into 23 series, and analysed their user agreements and privacy policy content to explore their ethical guiding role in academic publishing. The survey study found that LLMs user agreements had relevant provisions on intellectual property, data privacy and security, quality and integrity, and so on, but there were still controversial and needed to be further improved contents, such as the ownership of LLMs output, and continuous supervision and update of user agreements. The publishing community should collaborate closely with academics, technical developers, and legal experts to promote the formation of an ethical framework for the application of generative AI technology in academic publishing. This study systematically analyses the guidance of user agreements for LLMs in academic publishing ethics, filling a gap in the literature on ethical norms in the field of academic publishing for generative AI.

Keywords

Generative artificial intelligence; large language models; publication ethics; user agreements

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1. Introduction

Generative artificial intelligence (AI) refers to algorithms that can be used to create new content, such as text, code, images, videos and audios [1]. In 2018, the American OpenAI company released the GPT (Generative Pre-trained Transformer) large language model (LLM), which is trained by an extensive corpus of text and can generate human-like natural language texts [2]. OpenAI has successively released more sophisticated models, such as GPT-2 in 2019, GPT-3 in 2020 and GPT-4 in 2023, which have higher text generation quality and broader language understanding capabilities [3]. It is also the first time that nonhuman (ChatGPT) appeared on the list of top ten scientists in 2023 published by Nature [4]. A large number of companies and research institutions began to develop generative AI models, such as Google, Meta, Baidu and Alibaba.

With the advent of LLMs such as ChatGPT, the status of generative AI in academic research and publication has changed dramatically. Since LLMs have excellent language understanding, generation, and knowledge reasoning capabilities, and these models are easily accessible to the public, the application of LLMs in scientific publishing is rapidly expanding, including paper generation, literature summarisation, review, etc [5–8]. Liang et al. quantitatively compared GPT-4 feedback with human peer reviewers. They found that the overlap in the points raised by GPT-4 and by human reviewers was comparable to the overlap between two human reviewers, most of the users thought it was more beneficial than at least some human reviewer's feedback [9]. In addition, ChatGPT can generate human-like 'scientific papers' despite the data being entirely generated [10,11].

Generative AI may revolutionise research practices and publication, with opportunities and risks coexisting [12–14]. While accelerating the research and publication process, it also brings new ethical challenges, especially in academic integrity, copyright protection, data privacy, and attribution of responsibility. Due to the inherent technical limitations of generative AI, the content generated by ChatGPT or other LLMs may be erroneous, distort scientific facts, and spread false information [15,16]. The training data for LLMs mainly come from the Internet, although they have been screened and filtered before being input to remove low-quality and incorrect content, these data will not be truly neutral or objective due to the large amount of data and the limitations of current filtering technologies [16]. The use of LLMs may lead to information leakage, even if the input information does not contain private information, LLMs have the ability to infer sensitive information based on the training data. Thus, it is important for all stakeholders in the research process to implement this technology ethically and transparently. A scoping review that included 33 papers on recommendations for using LLMs in academia summarised five pieces of advice: (1) LLMs cannot be listed as authors in academic papers; (2) if considering using LLMs in academic work, authors should at least have a basic understanding of LLMs; (3) do not use LLMs to write the entire content of a paper; humans must be responsible for the use of LLMs, and the content generated by LLMs should be carefully verified by humans; (4) LLMs can be used for editing and optimisation of text; (5) any behaviour using LLMs should be transparent and clearly stated in the paper [17].

The user agreement or terms of services is a contract between the Internet service provider and the user on the provision and acceptance of Internet services, which is drafted by the provider, and the user reaches 'a mutual expression of will' with the provider by clicking on the 'agree' or 'read' option. The agreement is equally binding on both parties after it is signed. The Internet user agreement is an important agreement for Internet service providers to stipulate the rights and obligations of network users. The user agreements of LLMs usually stipulate the copyright ownership of user's content and content generated by LLMs, as well as measures taken by developers to ensure data security and others. Therefore, the user agreement of LLMs plays a crucial role in academic publishing, ensuring that the above principles are implemented. Yet, there is currently no research on the potential ethical guidance role of LLMs user agreements in the use of LLMs in academic publishing. The present study sets out to comprehensively explore the ethical implications of generative AI in academic publishing, with a specific focus on the role of user agreements. As such, this study addresses the research questions: How do user agreements of LLMs influence ethical practices in academic publishing, and what improvements are needed to ensure responsible use of generative AI technologies in this context?

2. Methods

It has been reported that the number of AI large language models worldwide has exceeded 1,328 (https://en.ce.cn/main/latest/202407/03/t20240703_39058393.shtml). In this study, we selected the LLMs on the LLMs evaluating list provided by China Telecom Research Institute (CTRI) combined with Shanghai Artificial Intelligence Laboratory, Tsinghua University and other institutions (34 models, https://mp.weixin.qq.com/s/Vbv0XD_vSltpE_-nKQDxtQ, accessed on 1 September 2024), and LLMs leaderboard provided by Artificial Analysis (64 models, <https://artificialanalysis.ai/leaderboards/models>, accessed on 1 September 2024) as our research samples. Although they may not fully capture the diversity of user agreements across the models, these LLMs included in the study present top models with high quality,

Table 1. Samples of large language models.

Models	Provider	Country	Provider founded time	Web site of user agreement	Update date	Previous version	Updates ^a	Model training ^b
Yi	01.AI	China	2023	https://platform.lingyiwanwu.com/useragreement	11 March 2024	No	NA	Yes
Qwen	Alibaba	China	1999	https://terms.alicdn.com/legal-agreement/terms/c_end_product_protocol/	No related updates	Yes		
Claude	20231011201348415/20231011201348415.html Anthropic	6 August 2024	10 June 2024	No related updates	Yes			
	Anthropic	United States	2021	https://www.anthropic.com/legal/consumer-terms	13 June 2024	1 May 2024	No related updates	Yes (not on any Materials which are not publicly available) Yes
Baichuan	Baichuan AI	China	2023	https://www.baichuan-ai.com/article/user-agreement	31 August 2023	8 August 2023	Authorisation mode for user's content was changed from 'worldwide, perpetual, irrevocable, free' to 'worldwide, free'	Yes
ERNIE	Baidu	China	2000	https://jyuan.baidu.com/info/User	1 August 2024	No	NA	Yes
Doubao	ByteDance	China	2012	https://www.doubao.com/legal/terms	No	No	NA	Yes
Moonshot	Moonshot AI	China	2012	https://login.moonshot.cn/user/argeement/modeluse	No	No	NA	Yes
DBRX	Databricks	United States	2013	https://www.databricks.com/legal/open-model-license	27 March 2024	No	NA	Unknown

(continued)

Table 1. Continued

Models	Provider	Country	Provider founded time	Web site of user agreement	Update date	Previous version	Updates ^a	Model training ^b
DeepSeek	DeepSeek	China	2023	https://chat.deepseek.com/downloads/DeepSeek%E7%94%A8%E6%88%B7%E5%8D%8F%E8%AE.html	30 August 2024	No	NA	Yes
Gemini	Google	United States	1998	https://ai.google.dev/gemini-api/terms?hl=zh-cn	14 May 2024	No	NA	Yes (users can choose to opt out; data for paid services is not used for training) Unknown
Gemma	Google	United States	1998	https://ai.google.dev/gemma/terms	1 April 2024	21 February 2024	No related updates	Unknown
Llama	Meta	United States	2004	https://www.facebook.com/legal/ai-terms	23 July 2024	No	NA	Yes
MiniMax-abab	MiniMax	China	2021	https://platform.minimaxi.com/protocol/user-agreement	No	No	NA	Yes
Mistral	Mistral AI	France	2023	https://mistral.ai/terms/#terms-of-service-la-plateforme	No	No	NA	Yes (users with paid services have the option to opt out of the training data) Yes (users can choose to opt out)
GPT	OpenAI	United States	2015	https://openai.com/policies/terms-of-use/	31 January 2024	14 March 2023	'use non-API Content to help improve our Services' was change to 'use Content to provide, maintain, develop, and improve our Services'	

(continued)

Table 1. Continued

Models	Provider	Country	Provider founded time	Web site of user agreement	Update date	Previous version	Updates ^a	Model training ^b
SenseChat	SenseTime	China	2014	https://chat.sensetime.com/wb/policy?id=1&lang=zh_CN	15 April 2024	No	NA	Yes
ChatGLM	Zhipu AI	China	2019	https://chatglm.cn/agreement?lang=zh	29 January 2024	No	NA	Yes
InternLM	Shanghai Artificial Intelligence Laboratory	China	2020	https://internlm-chat.intern-ai.org.cn/protocol/service	27 September 2023	No	NA	Unknown
Command	Cohere	Canada	2019	https://docs.cohere.com/docs/c4ai-acceptable-use-policy ; https://cohere.com/terms-of-use	7 September 2022	No	NA	Yes
Sonar	Perplexity	United States	2022	https://www.perplexity.ai/hub/legal/terms-of-service	4 June 2024	No	NA	Yes
Phi-3	Microsoft Azure	United States	2010	https://www.microsoft.com/licensing/terms/product/ForOnlineServices/all	1 September 2024	1 August 2024	No related updates	No
Reka	Reka	United States	2022	https://www.reka.ai/terms-of-use	January 2024	No	NA	Yes
Jamba	AI21 Labs	Israeli	2017	https://studio.ai21.com/terms-of-use	3 August 2023	No	NA	Yes

^aUpdates refer to updated content related to copyright of Materials, privacy policy, and model training.

^bModel training refers to whether the Materials are used for model training; NA: not applicable.

Table 2. Licence for user's content.

Models	Ownership	NE	WW	RF	P	T/S	NT/S	I
Yi	user			✓				
Qwen	user		✓	✓	✓	✓		
Claude	user			✓				
Baichuan	user		✓	✓				
ERNIE	user			✓				
Doubao	user		✓	✓	✓	✓		
Moonshot	original author	✓	✓	✓	✓	✓		
DBRX	user							
DeepSeek	user							
Gemini	user	✓	✓	✓		✓		
Gemma	user							
Llama	user		✓	✓		✓		
MiniMax-abab	user	✓				✓		
Mistral	user	✓	✓	✓			✓	✓
GPT	user			✓				
SenseChat	user		✓		✓	✓		
ChatGLM	original author	✓	✓	✓				✓
InternLM	user	✓	✓		✓	✓		✓
Command	user	✓	✓			✓		✓
Sonar	user		✓	✓		✓		✓
Phi-3	user							
Reka	user	✓	✓	✓	✓	✓		✓
Jamba	user	✓	✓	✓		✓		✓

NE: non-exclusive licence; WW: worldwide licence; RF: royalty-free licence; P: perpetual licence; T/S: transferrable/sublicensable licence; NT/S: non-transferrable/non-sublicensable licence; I: irrevocable licence.

performance and security, which are critical to user engagement. CTRI conducted the comprehensive assessment of international mainstream LLMs from four aspects, including capability, task, performance, and security. Artificial Analysis developed a leaderboard evaluating quality and performance across > 100 serverless LLM API endpoints. The performance benchmarking measures the end-to-end performance experienced by customers of LLM inference services, which means that the results are intended to represent the real-world performance customers experience across providers. Some of the LLMs in the two lists are overlapping. These models include various versions of a certain model, and they use the same user agreement and privacy policy, so we grouped them into a series, for instance, GPT-4, GPT-4 turbo, and GPT-3.5 were grouped into GPT.

We retrieved the official website of the models to collect their user agreement and privacy policy, and the embedded documents. Then we extracted the basic information of LLMs and information related to intellectual property and data privacy, including the update date of the user agreement, the ownership and licence types of the copyright of user's content and the output, whether the user's content was used to train the model, and others. The retrieval time was September 2024. The 'user's content' in this paper referred to the information or content provided by users to the LLMs through posting, publishing, uploading and other ways when they use the model, including but not limited to user registration or authentication information, pictures, data, text, sound, etc. The 'output' referred to all the content generated by LLMs according to user's prompts or uploaded content. The user's content and output were collectively called 'Materials'.

3. Results

3.1. General characteristics of selected LLMs

The 98 LLMs were divided into 23 series offered by 22 providers from five countries (see Table 1), and the number of LLMs indicated in the study refers to the number of series hereafter. Seven (31.8%) provider companies were established after 2020. Eight LLMs are not commercially available, among which InternLM and Baichuan can be commercially used by application. Seven LLMs are commercially available, and the other eight LLMs are not specified as being commercially available or not. There are 19 LLMs indicating that the Materials will be used for training the model, where GPT, Gemini, and Mistral (paid services) allow users to choose content not to be used for training the model. Claude specifies that content that is not publicly available will not be used for training the model. Gemini also indicates that the content is

Table 3. Licence for output.

Models	Ownership	NE	WW	RF	P	T/S	NT/S	I	R	A
Yi	user			✓						
Qwen	user		✓	✓	✓	✓				
Claude	provider									✓
Baichuan	provider	✓					✓		✓	
ERNIE	provider	✓					✓		✓	
Doubao	user		✓	✓	✓	✓				
Moonshot	user	✓	✓	✓	✓	✓				
DBRX	user									
DeepSeek	user									
Gemini	user	✓	✓	✓		✓				
Gemma	user									
Llama	provider									
MiniMax-abab	provider									
Mistral	user	✓	✓	✓			✓	✓		
GPT	provider									✓
SenseChat	provider									
ChatGLM	user	✓	✓	✓				✓		
InternLM	user	✓	✓		✓	✓		✓		
Command	provider	✓					✓		✓	
Sonar	provider	✓					✓			
Phi-3	user									
Reka	user									
Jamba	provider	✓					✓			

NE: non-exclusive licence; WW: worldwide licence; RF: royalty-free licence; P: perpetual licence; T/S: transferrable/sublicensable licence; NT/S: non-transferrable/non-sublicensable licence; I: irrevocable licence; R: revocable licence; A: assignment.

not for model training in the paid services. Seven LLMs state that the privacy data will be de-identified or anonymized before being used for model training.

3.2. General characteristics of the user agreement

The user agreements investigated in this study are mainly located on the website page in the following forms: autopopup, checkable with link, and default. Autopopup mainly refers to that after users click ‘Sign Up’, the user agreement will pop up automatically on the page, and users can only proceed to the next step after clicking agree. Checkable with link mainly refers to that users can check whether they agree after reading the content linked. With the form of default, the platform only attaches the user agreement to the registration service page without any operation such as ‘check’ and ‘agree’, and this mode has the least reminder effect. Formally, the user agreements investigated in this study adopt a list form, which is hierarchical, itemised and clearly and orderly present the terms content. In addition, LLMs mostly choose to use font bolding or underlining processing to remind users of important content, such as privacy and licence terms. Yet, there are also LLMs which do not take any measures to make relevant terms easy to perceive, such as Baichuan model provided by Baichuan AI. Nineteen models provided the update time of the user agreement, and only six models provided the previous version, among which Baichuan updated the licence type for user’s content, and GPT updated its data scope for model training (see Table 1).

3.3. Copyright terms for user’s content

All the models explicitly state that users retain ownership in the input. The exceptions are Moonshot and ChatGLM, which claim that the copyright of any content users input into the LLMs belongs to the original author. Also, they all emphasise that users should undertake that the input dose not infringe upon any legal rights of third parties. The LLMs require a licence rather than a transfer of copyright. When a user uploads content to the LLMs, they are typically granting permission for the LLMs to use that work.

The majority of these LLMs include specific licence and use rights provisions, and the licences are usually stated in a string of legal terms. For example, Mistral requires that users grant them a ‘worldwide, non-revocable, non-exclusive, non-sublicensable, non-transferable, royalty-free right to use the Prompts, Outputs and Input Data for the purpose of

performing the Services'. Another four LLMs do not have any information about licence terms, and these LLMs were greyed out in Table 2. It should be noted that the absence of licence terms does not mean that the LLMs lack the rights to use the content. An implied copyright licence can be created by law in the absence of an actual agreement between the parties, which have been used to grant licences in situations where a copyrighted work was created by one party at the request of another [18]. However, for this study we focus on the licences as specifically stated, that is, those that we know the user grants by agreeing to a user agreement or terms of service.

The common licence types for user's content include non-exclusive, worldwide, royalty-free, transferable, sublicensable, perpetual and irrevocable. Non-exclusive (nine LLMs) means that the user can use the content and grant the licence to others. Worldwide (14 LLMs) means the licence is effective all over the world. Royalty-free (15 LLMs) means that the provider can use the content for free. Sublicensable (12 LLMs) means that the provider can give the intellectual property rights to a third party without direct permission from the user. Most licences are sublicensable and transferable which affords the provider the option to perform either. Perpetual (six LLMs) means that the licence authorises the provider to use the content indefinitely. Irrevocable (7 LLMs) means that the licence can only be terminated for reasons specified under the termination provisions [19] (see Table 2).

3.4. Copyright terms for output

As to the ownership of output, 13 LLMs state that users own the output, and 10 LLMs point out that the provider retains the ownership of the output. In our dataset, the types of licences for the output include revocable and assignment besides the types as for user's content. GPT and Claude assign the right to the users. For example, the terms of use for GPT describes this as '*We hereby assign to you all our right, title, and interest, if any, in and to Output*'. In addition, the LLMs which own the copyright of output tend to grant a non-transferable/non-sublicensable and revocable licence to users, while those in which users own the right prefer to require a transferable/sublicensable licence for the output. Eight LLMs do not provide a specific statement of the licence types, and these LLMs were greyed out in Table 3.

4. Discussion

Formally, the user agreements or terms of services belong to a format contract, and LLMs platforms as providers of the contracts should fulfil their reasonable reminding obligations for the important terms. The reasonable reminding of electronic contracts should be easy to obtain. It has been pointed out that paraphrased user agreement which display content using simple terms across multiple windows are more probably to elicit positive attitudes and behaviours among prospective users [20]. From this aspect, most of the LLMs user agreements investigated in this study present the agreement content well. From the perspective of fairness, some LLMs require users to grant the provider an unlimited licence to use user's content, such as a perpetual, worldwide or sub-licensable licence right, while users generally only get limited use of the application after agreeing to the agreement, and if there is any loss caused by user's content to the provider, users need to bear relevant legal liabilities and compensate for losses. Therefore, the relevant requirements of the provider have obvious asymmetry with respect to the rights enjoyed by users. In addition, according to the user agreement, users only have two choices – agreeing to the terms of the agreement or abandoning related services, which will undoubtedly increase the unfairness of the contract. This indicates that there is much room for optimisation in the terms rationality.

The user's content mainly includes original content, personal information and other data. The LLMs user agreement investigated in this study stipulates that users should ensure that they have or have obtained sufficient, necessary and effective legal permission and authorisation from the right subject involved in the uploaded content, and do not infringe the legitimate rights and interests of any third party. LLMs may use the user-uploaded content for model training, which improves the performance and quality of service of the model, but the model may provide the content to other users. Application of LLMs to reviewing papers may result in important data leakage, and ultimately infringe the intellectual property rights of the original authors [15]. Therefore, there is a very high risk of intellectual property rights infringement by using LLMs for peer reviewing. Reviewers should confirm that the author agrees to use LLMs to assist in reviewing before doing that. Claude provided by Anthropic stated that 'We will not train our models on any Materials that are not publicly available', and GPT, Gemini and Mistral (paid services) allow users to choose to opt out of training data. This to some extent avoids the leakage of research data of original authors, other LLMs can refer to this strategy to enhance user's trust in model services and improve overall user experience.

We found in the study that 10 (43.5%) LLMs specify that the ownership of content generated by LLMs belongs to the provider, and 13 (56.5%) LLMs indicate that users retain the ownership of the output. Actually, the question of who owns the output is much complicated. At present, there is still no consensus in the academic community on the ownership of the content generated by LLMs: users, LLMs providers, LLM developers or owners of training data [15]?

Besides, whether the generated content is protected by copyright is also under discussion. There was a case in 2022, where the United States Copyright Office (USCO) approved a graphic novel by Kris Kashtanova utilising AI-generated art, as an intellectual property protected by copyright. However, subsequent amendments to the copyright certificate have excluded the AI-generated images contained in the book, stating that ‘AI-generated images cannot receive copyright protection if they are made with little human input’, but there were no other copyright protection decisions for outputs such as videos and codes [21]. On the other hand, a trained LLM, ideally, will produce completely new content. However, it is possible that text generated by LLMs may be very similar to existing text, and sometimes even almost identical. This raises the question of whether the generated work infringes upon the copyright of the original owner. This could result in copyright infringement liability for the creators of the model and for users who use the generated work. In fact, users may be liable for infringement even if they had no knowledge of the original work [22].

In addition to content copyright, some LLMs indicate in the user agreement that they will add AI-generated identifiers to generated content, including explicit watermark identifiers that are visible to the naked eye on the generated content and implicit watermark identifiers that cannot be directly perceived by humans but can be extracted from the content through technical means; some remind users to mark the generated content as AI-generated in a prominent way. These provisions to some extent cater to the current mainstream policy on transparent use of AI. In addition, user agreements mentioned that the output of models is not always correct, and users need to actively check the accuracy and authenticity of the output content. Studies have found that AI can generate fraudulent but authentic-looking medical papers with obvious errors in the paper itself and especially in the references [23–27]. Similarly, in peer review, although LLMs can summarise the article content well, it cannot provide specifically critical and instructive suggestions [28]. What’s worse, these suggestions might mislead the decision by editors. Therefore, although LLMs are a powerful tool that can help at multiple stages of academic publishing, authors, reviewers, and journals should add human verification steps to relevant content, retain vigilant or even prohibit the use of this technology at certain stages. For example, Korean Journal of Radiology prohibits reviewers from using LLMs to generate review comments, and LLMs can only be used for language polishing, including modifying grammatical and spelling errors, adjusting formats, etc. [29]. AI-generated images and other multimedia are not permitted in the Science journals without explicit permission from the editors, and reviewers may not use AI technology in generating or writing their reviews because this could breach the confidentiality of the manuscript [30].

Although there are some tools to detect AI-generated content at present [31,32], the effectiveness of detection tools may also be limited due to the continuous iterative updates of AI-based tools. Therefore, instead of a futile competition between AI tools and AI detectors, it is better to encourage or require transparency in the use of AI, such as making raw data publicly available, improving the quality of peer review, strengthening the scrutiny, etc. [33]. The ‘Guideline on the Boundaries of AIGC usage in Academic Publishing’ was officially released in September 2023 [34]. It elaborates on the principles of using AIGC in academic publishing, including transparency and accountability, quality and integrity, privacy and security, fairness, sustainable development, among which transparency and accountability are the most basic principles. It also introduces behavioural frameworks and practical guidelines for authors, research institutions, academic journal publishers. Committee on Publication Ethics (COPE) [35], International Committee of Medical Journal Editors (ICMJE) [36] and the World Association of Medical Editors [37] also explicitly require that if authors use any AI tools to write articles, they should be transparent and need to provide relevant information about the tools and their use in appropriate places such as the materials and methods section; authors are responsible for the accuracy and completeness of AI-generated content.

User agreements of LLMs play an important role in promoting these principles being implemented. First, user agreements provide users with behavioural guidance to comply with established moral and legal standards when using this tool, such as requiring users to input content that does not infringe third-party rights. This to a certain extent can prevent potential academic misconduct such as plagiarism and data fabrication. Second, user agreements require users to make appropriate disclosures when using the tool so that others can clearly understand the degree of participation of AI tools, which helps to improve the transparency of AI tool use. Third, according to the results of this survey, user agreements also require users to review and verify the output of LLMs to ensure its accuracy and reliability, which helps to guarantee the high standard of academic research and prevent the spread of misleading or inaccurate information. Fourth, privacy and security are basic legal principles for the application of AIGC in academic publishing. On one hand, the user agreement requires users to avoid uploading private data as much as possible; on the other hand, it also requires developers to adopt appropriate technical measures to protect privacy and data security. Fifth, Since AI has the risk of replication and amplification bias, developers should be required to carefully evaluate and review potential sources of bias during training data selection, algorithm design, model generation and optimisation processes in user agreements and make corresponding optimisations to reduce bias. Finally, most user agreement give a statement of the ownership and licence terms

of Materials, and whether the Materials are used for model training, users should carefully read these terms before using although some of the issues need to be further discussed.

To facilitate the implementation of the proposed ethical frameworks, we offer the following actionable recommendations. First, we propose the establishment of a multidisciplinary task force. A collaborative task force comprising academic publishers, AI developers, legal experts, and ethicists should be formed to develop and improve the regulatory system of user agreements. Operating under the guidance of established organisations such as COPE or ICMJE, this task force would ensure alignment with current ethical standards and foster interdisciplinary consensus. Second, we recommend the development of a standardised template for user agreements. A globally adaptable template for user agreements should be created, incorporating mandatory provisions on intellectual property rights, data privacy, transparency in AI-generated content, and dispute resolution mechanisms. Third, we emphasise the importance of enhanced education and training initiatives. Comprehensive training programmes and resources should be developed to educate researchers, authors, editors, reviewers and publishers on the ethical use of generative AI tools. These initiatives would raise awareness of ethical guidelines and promote best practices, empowering stakeholders to navigate the complexities of AI-driven academic publishing responsibly. Finally, we advocate for a dynamic approach to ethical frameworks, where user agreements and guidelines are regularly reviewed and updated to reflect technological advancements and evolving ethical considerations.

5. Conclusion

This study has systematically analysed the critical role of LLM user agreements in shaping ethical practices within academic publishing, specifically focusing on intellectual property, data privacy and security, quality and integrity. Our primary contribution lies in being one of the first empirical analyses to demonstrate how these often-overlooked legal documents serve as foundational, yet somewhat ambiguous, guidance for generative AI ethics in the scholarly domain. We have identified specific clauses within current agreements that create substantial uncertainty, particularly regarding the ownership of AI-generated content and the mechanisms for ongoing oversight and updates. This ambiguity poses tangible risks to core academic values.

In spite of the important findings from the user agreements of LLMs, we have to acknowledge the limitations of this study. First, our analysis is primarily based on the publicly available user agreements of a select group of major LLM providers, such as ChatGPT, Gemini, and Claude. It may not fully capture the diversity of agreements from emerging models. Second, the research relies heavily on textual analysis of the agreements themselves. It does not empirically investigate how these agreements are interpreted and implemented by key stakeholders (authors, reviewers, editors, publishers) in real-world publishing workflows, nor does it measure their actual impact on publishing outcomes.

Based on the significant findings and limitations of this study, future research can conduct qualitative and quantitative studies, such as surveys and interviews, to understand how researchers, editors, publishers, and AI developers perceive, interpret, and navigate the ethical challenges posed by LLM user agreements in their work. In addition, future research should investigate the actual impact of user agreement stipulations and evolving AI capabilities on specific publishing practices, such as peer review, authorship attribution, and plagiarism detection. Furthermore, developing and rigorously testing the effectiveness, feasibility, and adaptability of proposed ethical guidelines and governance models for AI in academic publishing is another important direction.

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